

Seydi Cheikh Wade

+1 (514) 431-3541 | seydiwade@cmail.carleton.ca | [linkedin.com/in/seydi-c](https://www.linkedin.com/in/seydi-c) | seydiwade.wiki

EDUCATION

Carleton University

B.Eng. Software Engineering (Co-op) – GPA: 3.6/4.0

Ottawa, ON

Graduating May 2028

- Coursework: Database Systems, Algorithms & Data Structures, Operating Systems, OO Software Development, Real-Time Concurrent Systems, Requirements Engineering, Computer Architecture

WORK EXPERIENCE

Statistical Developer (Co-op)

May 2026 – Present

Statistics Canada

Ottawa, ON

- Built** and validated **50+ SAS analytical programs in R** for Statistics Canada's **Canadian Centre for Justice and Community Safety Statistics**, supporting statistical production for the nationwide **Uniform Crime Reporting (UCR) Survey**
- Engineered reusable **quality controls** for statistical production workflows, validating outputs from structure through final values and enforcing runtime checks for **source types, join-key uniqueness, and output integrity**
- Proposed and presented a **GitLab-based code-management strategy** to the Unit Head, replacing fragmented local/shared-folder workflows with centralized version control, branch-based collaboration, and structured issue tracking; secured **Centre-level approval** and now support implementation across the division

Junior Data Scientist (Co-op)

Jun. 2025 – Sep. 2025

Jasmine Conseil – Government Contractor

Laval, QC

- Built an **ML prediction system** for the Senegalese Customs agency, responsible for **40% of national budget operations**, to automate control circuit assignment across **10,000+ import declarations**
- Benchmarked 5+ classifiers; selected **MLPClassifier+PCA** achieving the highest **ROC-AUC (0.99)** and **F1-score (0.86)** among all evaluated models on imbalanced data; serialized with Joblib for production serving
- Deployed the trained model as a **FastAPI** REST API on a **DigitalOcean Linux VM** with Pydantic validation and Swagger docs; configured **firewalld** rules and routed traffic through a **MuleSoft Anypoint proxy** for rate limiting

PROJECTS

ChizuCode – Hackathon Winner ↗ | Next.js, FastAPI, pgvector, Gemini, Voyage AI

- Won MLH Best Use of Gemini API** at ConHacks 2026 (team of 4); built a **GitHub repo visualizer** with Gemini summaries, **pgvector** embeddings, and source-grounded **RAG Q&A**
- Implemented **RAG Q&A** with **pgvector** similarity search, relevance filtering, and source-grounded **Gemini** responses over repository content

Soki – Hackathon Winner ↗ | Next.js, FastAPI, Gemini, FSRS

- Won 1st place** (AI Automation) and **runner-up** (Best Language Cognition) at Carleton MindHack; built a **Gemini-powered study platform** with PDF-to-flashcard generation and **FSRS** scheduling

Fallback – Real-Time Man-Down Detection ↗ | Python, PyTorch, FastAPI, Next.js, SSE

- Built a real-time **deep-learning man-down detector** with **PyTorch** pose estimation, then integrated alerts, annotated frames, and **A* routing** through a **FastAPI/SSE** backend and **Next.js** dashboard

EXTRACURRICULAR & VOLUNTEERING

Software Engineer (Volunteer)

May 2026 – Aug. 2026

Develop for Good – Crisis Center, Inc.

San Francisco, CA (Remote)

- Rebuilt the public website from the ground up for Crisis Center, Inc. as part of a volunteer team, serving **30,000+ people annually** across Central Alabama; created clearer crisis-resource pathways with **14 reusable templates**, **6 filters**, and **Quick Exit safety controls**

TECHNICAL SKILLS

Languages: Python, R, SQL, Java, C, C++, JavaScript, TypeScript

ML & Data Science: PyTorch, deep learning, scikit-learn, pandas, NumPy, supervised learning (MLP, XGBoost, SVM, Decision Trees), PCA, feature engineering, F1-score/ROC-AUC optimization, imbalanced data handling

Frameworks & APIs: FastAPI, REST APIs, Pydantic, Joblib, Next.js, NextAuth, JDBC, Jupyter

Databases: PostgreSQL, pgvector

Developer Tools: Git, GitHub, Bitbucket, Docker, Linux CLI, Jira, Confluence

Cloud & Infrastructure: DigitalOcean, MuleSoft Anypoint